

Dhruv Karanwal

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LinkedIn



GitHub

SUMMARY

Third-year B.Tech Computer Science and Engineering (Artificial Intelligence) student at Vishwakarma Institute of Technology, Pune (CGPA: 8.84) with hands-on experience in Python, Flutter, and full-stack development across AI/ML and app development projects. Passionate about building real-world solutions at the intersection of AI and software engineering. General Secretary at IEEE VIT Pune, driving technical initiatives and team coordination.

EDUCATION

Vishwakarma Institute of Technology
Computer Science and Engineering (Artificial Intelligence)
CGPA: 8.84

Pune
2024-2028

WORK EXPERIENCE

App Developer Intern
Bandhavgarh Tiger Reserve

Pune
Dec 2025 – May 2026

- Developed a Flutter-based booking platform used for visitor registration and reservation management.
- Integrated Firebase for real-time synchronization and efficient booking workflows.
- Designed and implemented automated slot allocation and FCFS reservation mechanisms..

SKILLS

- Languages: C++, Python, SQL, Dart
- Frameworks & Libraries: FAISS, XGBoost, Sentence Transformers, OpenCV, Flutter, Flask, LangChain
- Developer Tools: Git, GitHub, Firebase, Google Colab, Jupyter Notebooks
- Databases: MySQL, Firebase Firestore

PROJECTS

AI-Powered Pre-Delinquency Intervention Engine:

Feb 2026 – May 2026

- Predicted loan default probability 2–4 weeks in advance by analyzing 30+ behavioral signals including salary shifts, credit utilization, and spending volatility using rolling 7/14/30-day trend windows.
- Built a Risk Fusion Engine combining Gradient Boosting and XGBoost outputs to generate a unified Financial Stress Index and classify customers into Low, Medium, High, and Critical risk tiers.
- Designed a personalized intervention layer triggering automated EMI flexibility, payment reminders, and escalation workflows based on real-time risk classification.

Sahyog – AI Disaster Relief Platform:

Nov 2025 – Jan 2026

- Built a tri-app disaster response system with AI-based SOS prioritization using risk severity, population density, and resource availability.
- Integrated Google Liquid Galaxy for immersive geospatial visualization of disaster zones with custom KML overlays

Neural Machine Reading – Multimodal Document QA:

Aug 2025 – Oct 2025

- Built a multimodal document QA system using OCR, AI layout segmentation, and Gemini for grounded answers with visual bounding-box citations.
- Implemented 5-stage hybrid retrieval fusing BM25, FAISS semantic search, and CLIP cross-modal search with weighted score fusion.

ACHIEVEMENTS

- National Hackathon Winner for developing Sahyog, an AI-powered multi-disaster relief and response platform.
- Patent Published: "Gen-AI Powered Healthcare Assistant for Chronic Disease Prediction and Medical Support", Intellectual Property India, 2026 (Application No: 202521113739).
- Organised a multi-day technical workshop on LLM for 200+ students.
- Department Representative (First Year) – Led and coordinated 1000+ students in inter-departmental competitions across multiple domains.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Joint Secretary
IEEE Student Branch, VIT Pune

June 2026 – Present

Department Representative
Vishwakarma Institute of Technology, Pune

Jan 2025 – June 2025